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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Student Name:Charis Fernandes Student Roll No: 26

Experiment – 9: Implement Containerization

Aim: To study and Implement Containerization using Docker on AWS/Azure/Google cloud platform.(Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to learn –
- How to implement containerization using docker

Lab objective mapped : ITDO8024.5 : To comprehend Docker and its role in the DevOps lifecycle.

Prerequisite: An AWS account

Requirements: Cloud Login**,** Desktop, Browser, Internet

Pre-Experiment Theory: (describe following topics in soft copy form)

1. **Amazon Elastic Container Service (Amazon ECS)**
   Amazon ECS is a fully managed container orchestration service provided by AWS. It allows you to **run, manage, and scale Docker containers** without needing to handle the underlying servers. ECS takes care of scheduling containers, maintaining their availability, and integrating with other AWS services. You can run it using **AWS Fargate (serverless)** or on your own EC2 instances, depending on how much control you want.

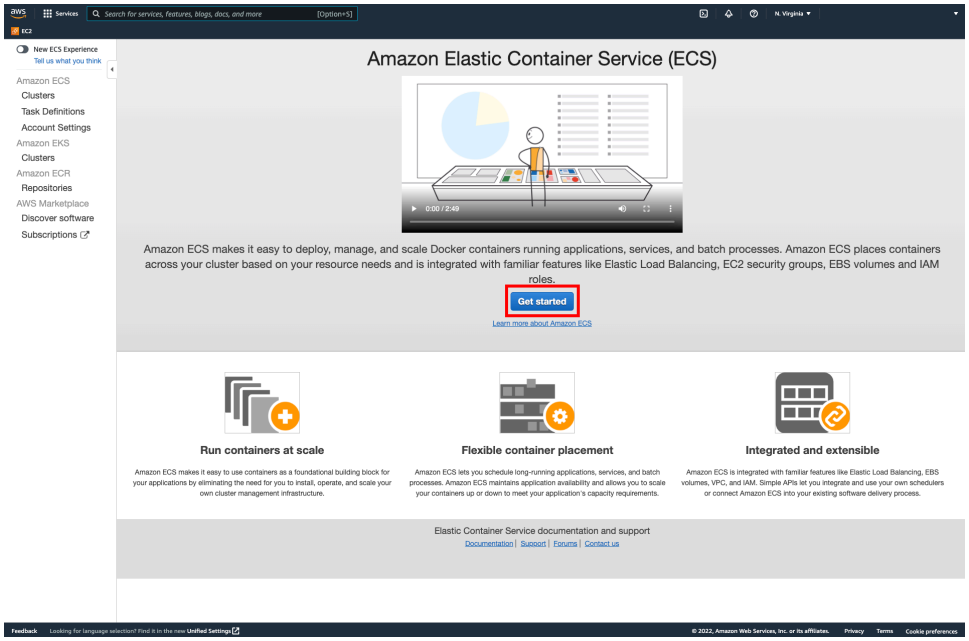
2. **Docker applications**
   Docker applications are software programs packaged into **containers**, which include the application code along with all its dependencies (libraries, runtime, system tools). This ensures the application runs consistently across different environments like development, testing, and production. Docker makes deployment faster, more portable, and less prone to “it works on my machine” issues.

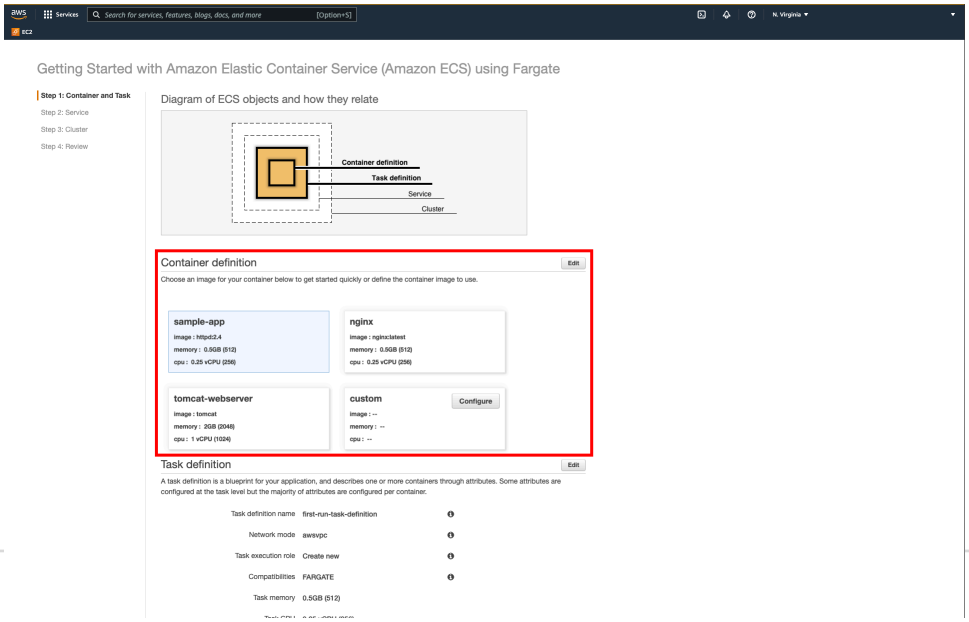
3. **Amazon ECS task definitions**
   A task definition in ECS acts as a **blueprint for running containers**. It defines important settings such as which Docker image to use, how much CPU and memory to allocate, which ports to open, and any environment variables needed. It can also define multiple containers that should run together. Every time you run a container in ECS, it uses a task definition to know exactly how to behave.

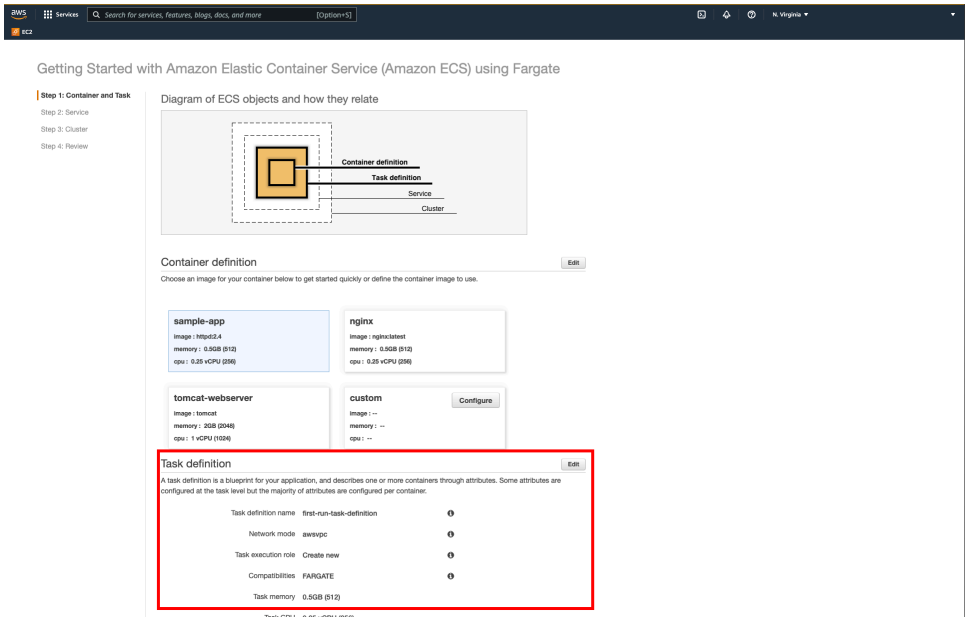
4. **Application Load Balancer**
   An Application Load Balancer (ALB) is responsible for **handling incoming traffic from users and distributing it across multiple running containers or services**. It improves availability and reliability by preventing any single container from being overloaded. ALB works at the application layer (HTTP/HTTPS) and can route traffic based on URL paths or domains. It also provides a **public DNS endpoint**, allowing users to access your application through a browser.

Procedure:

https://docs.aws.amazon.com/hands-on/latest/deploy-docker-containers/deploy-docker-containers.html






Getting Started with Amazon Elastic Container Service (Amazon ECS) using Fargate
Step 1: Container and Task
Step 2: Service
Step 3: Cluster
Step 4: Review
Diagram of ECS objects and how they relate
Container definition
Task definition
Service
Cluster
Container definition
Edit
Choose an image for your container below to get started quickly or define the container image to use.
sample-app
nginx
tomcat-webserver
custom
Configure
Task definition
Edit
A task definition is a blueprint for your application, and describes one or more containers through attributes. Some attributes are configured at the task level but the majority of attributes are configured per container.
Task definition name first-run-task-definition
Network mode awsvpc
Task execution role Create new
Compatibilities FARGATE
Task memory 0.5GB (512)

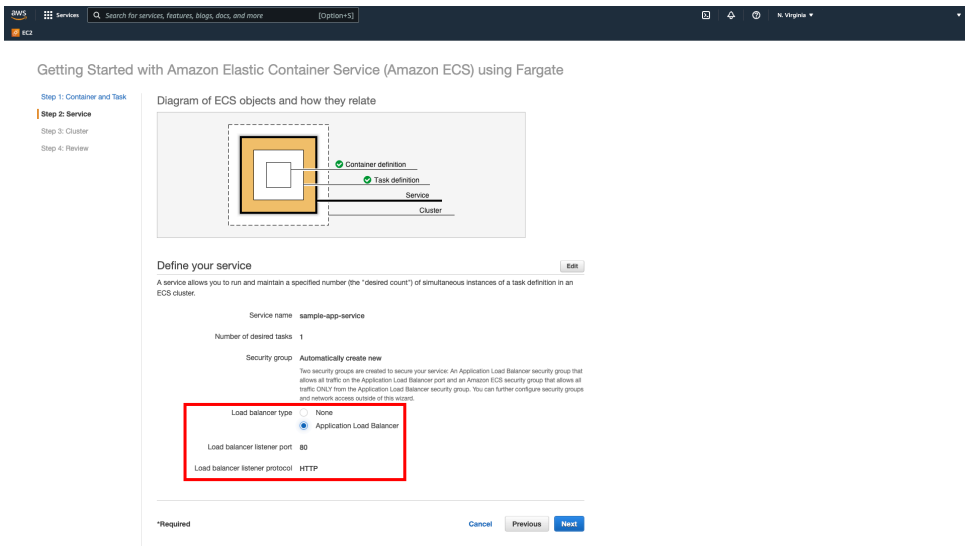

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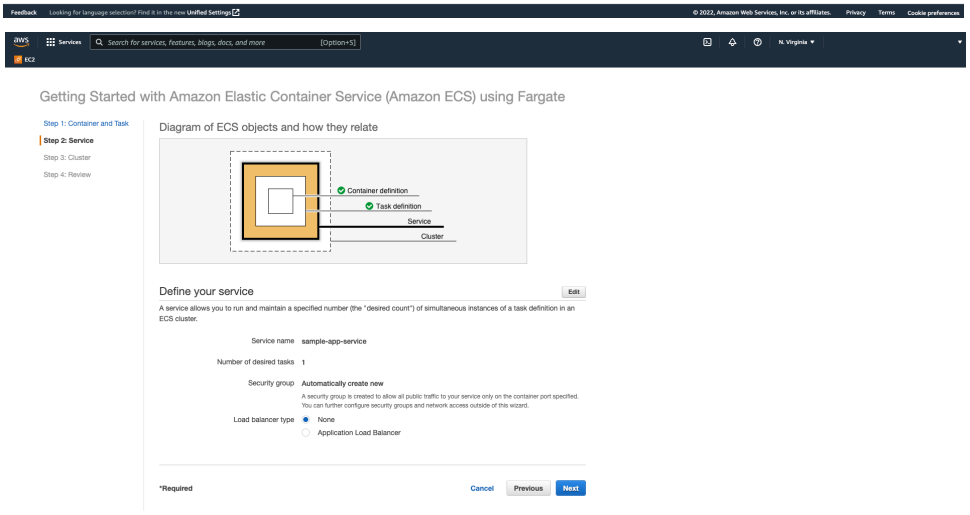
Diagram of ECS objects and how they relate
Container definition
Task definition
Service
Cluster
Define your service
Edit
A service allows you to run and maintain a specified number (the "desired count") of simultaneous instances of a task definition in an ECS cluster.
Service name sample-app-service
Number of desired tasks 1
Security group Automatically create new
Load balancer type None
Application Load Balancer
Load balancer listener port 80
Load balancer listener protocol HTTP
*Required
Cancel
Previous
Next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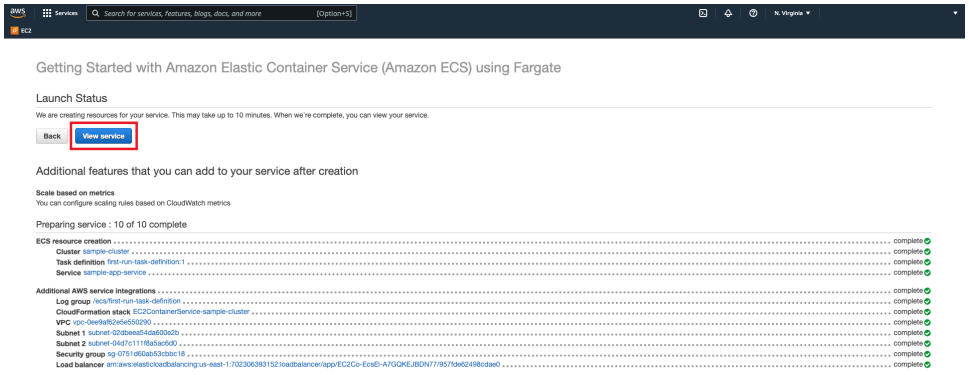
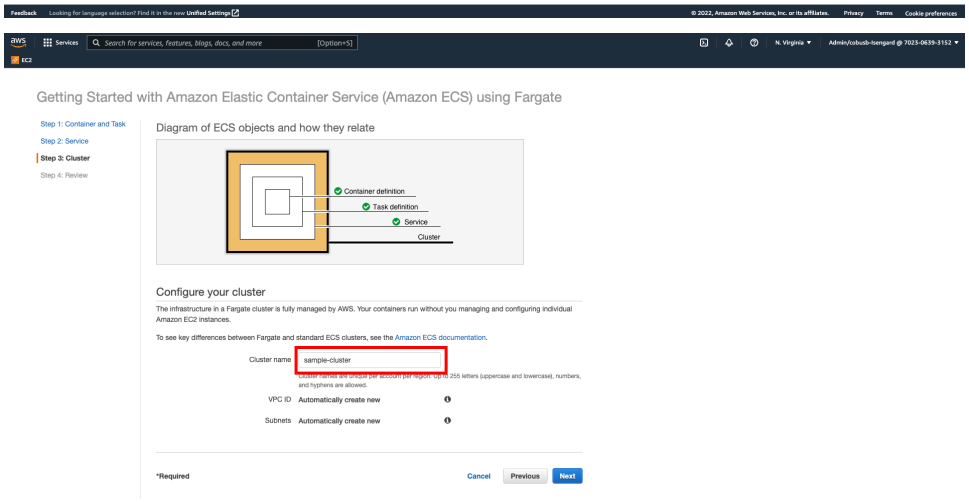
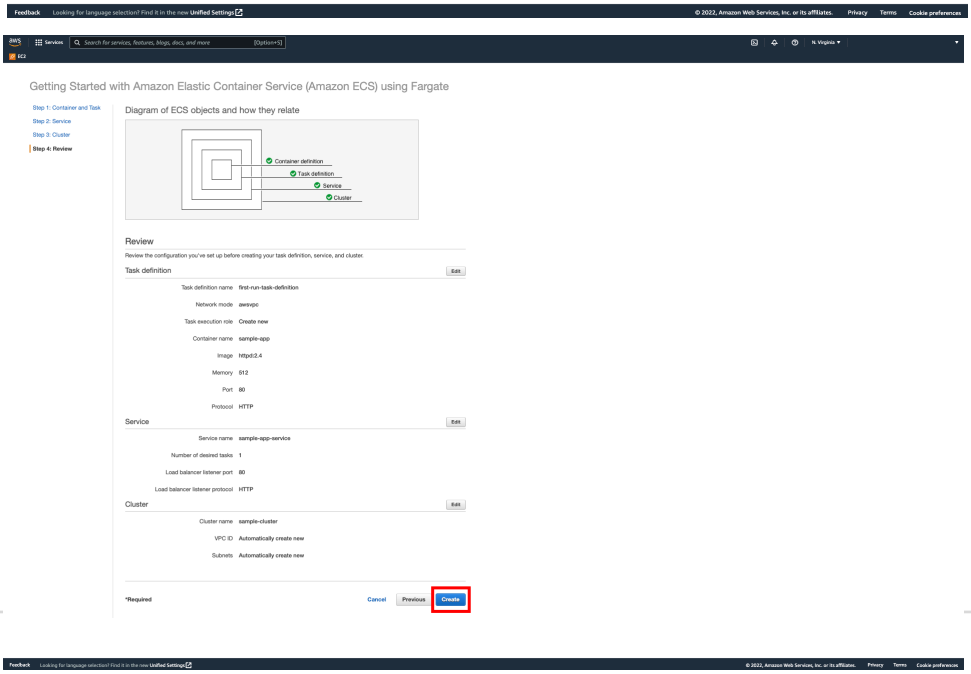
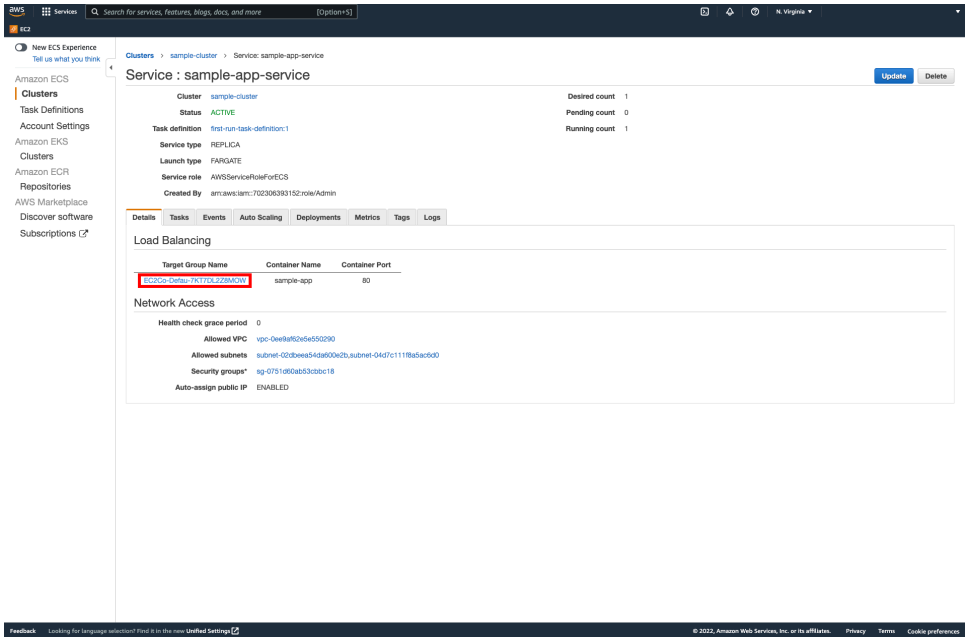
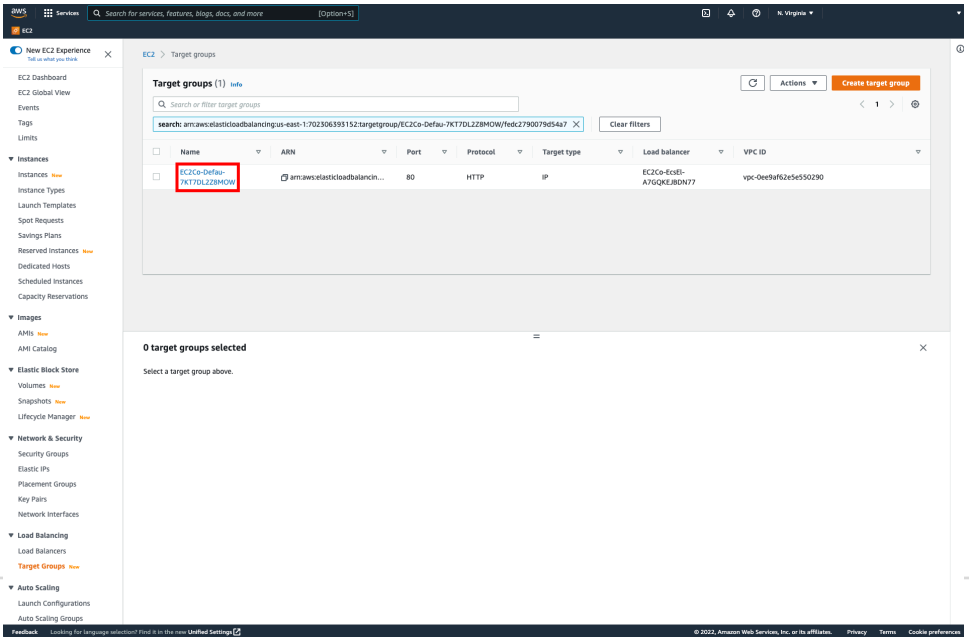
Container definition
Task definition
Service
Cluster
Define your service
Edit
A service allows you to run and maintain a specified number (the "desired count") of simultaneous instances of a task definition in an ECS cluster.
Service name sample-app-service
Number of desired tasks 1
Security group Automatically create new
A security group is created to allow all public traffic to your service only on the container port specified. You can further configure security groups and network access outside of this wizard.
Load balancer type None
Application Load Balancer
*Required
Cancel
Previous
Next

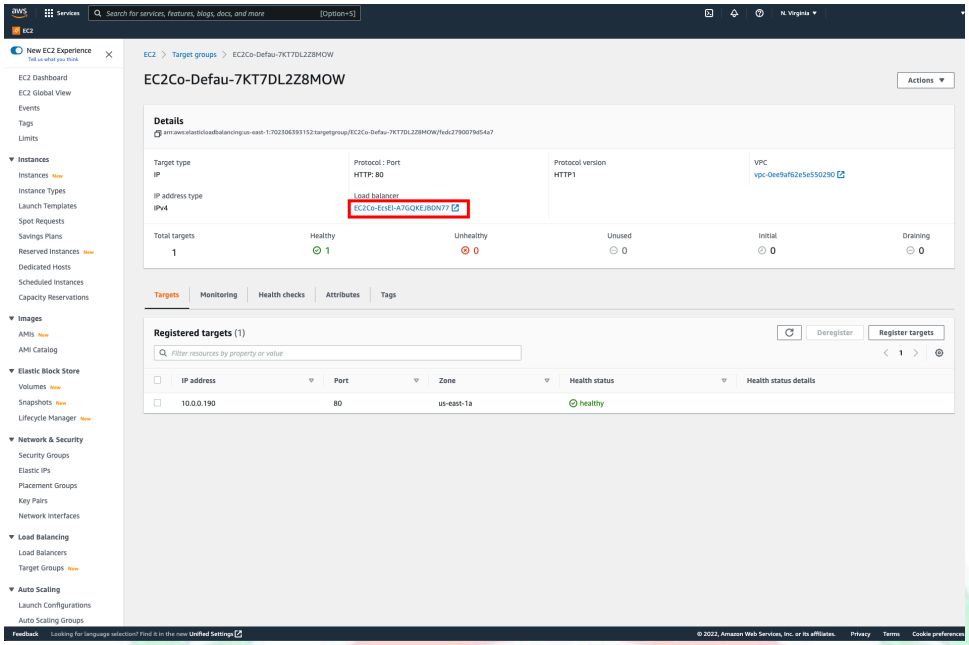

Getting Started with Amazon Elastic Container Service (Amazon ECS) using Fargate
Launch Status
We are creating resources for your service. This may take up to 10 minutes. When we're complete, you can view your service.
Back
View service
Additional features that you can add to your service after creation
Scale based on metrics
You can configure scaling rules based on CloudWatch metrics
Preparing service : 10 of 10 complete
ECS resource creation
Cluster sample-cluster
Task definition first-run-task-definition:1
Service sample-app-service
Additional AWS service integrations
Log group /ecs/first-run-task-definition
CloudFormation stack ECZContainerService-sample-cluster
VPC vpc-0ee9aff62e5e550290
Subnet 1 subnet-02dbeea54da60da2b
Subnet 2 subnet-04d7c111f8a5ac6d0
Security group sg-0751d60ab53cbbc18
Load balancer arn:aws:elasticloadbalancing:us-east-1:702306393152:loadbalancer/app/EC2Co-EcsEl-A7GQKEJBDN77/957fde62498cdae0
complete

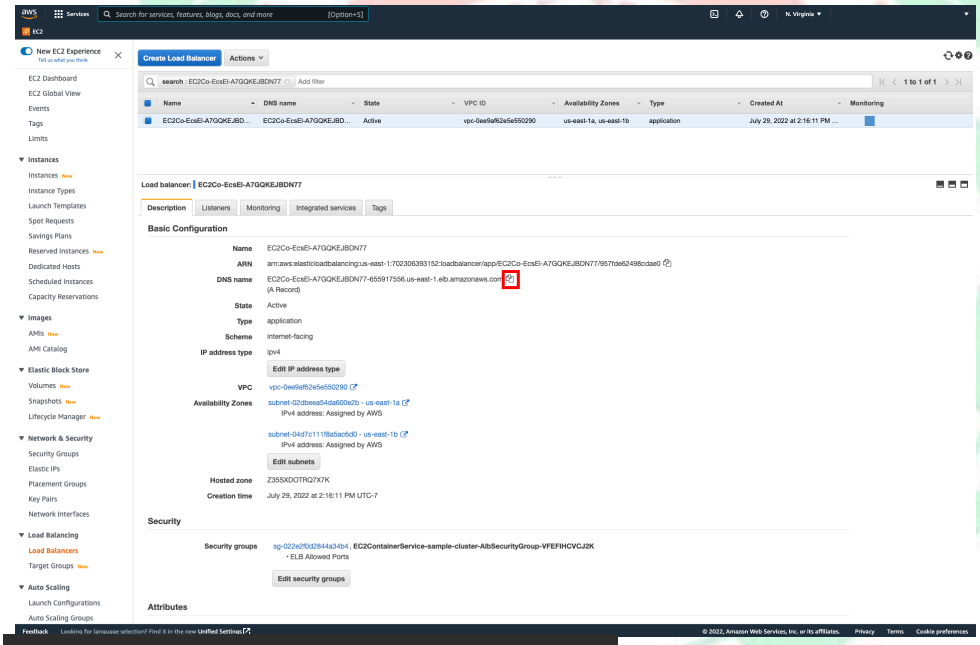

Getting Started with Amazon Elastic Container Service (Amazon ECS) using Fargate
Step 1: Container and Task
Step 2: Service
Step 3: Cluster
Step 4: Review
Diagram of ECS objects and how they relate
Container definition
Task definition
Service
Cluster
Configure your cluster
The infrastructure in a Fargate cluster is fully managed by AWS. Your containers run without you managing and configuring individual Amazon EC2 instances.
To see key differences between Fargate and standard ECS clusters, see the Amazon ECS documentation.
Cluster name
sample-cluster
VPC ID
Automatically create new
Subnets
Automatically create new
*Required
Cancel
Previous
Next

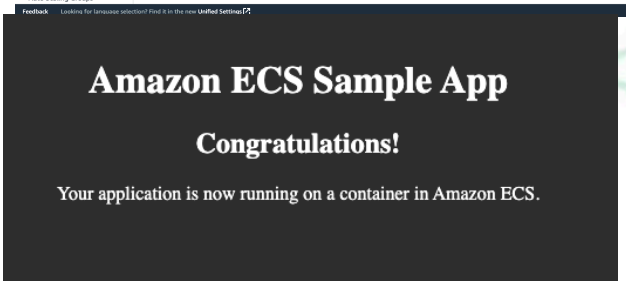

Getting Started with Amazon Elastic Container Service (Amazon ECS) using Fargate
Step 1: Container and Task
Step 2: Service
Step 3: Cluster
Step 4: Review
Diagram of ECS objects and how they relate
Review
Review the configuration you've set up before creating your task definition, service, and cluster.
Task definition
Edit
Task definition name first-run-task-definition
Network mode awsvpc
Task execution role Create new
Container name sample-app
Image httpd:2.4
Memory 512
Port 80
Protocol HTTP
Service
Edit
Service name sample-app-service
Number of desired tasks 1
Load balancer listener port 80
Load balancer listener protocol HTTP
Cluster
Edit
Cluster name sample-cluster
VPC ID Automatically create new
Subnets Automatically create new
*Required
Cancel
Previous
Create


Clusters > sample-cluster > Service: sample-app-service
Service : sample-app-service
Update
Delete
Cluster sample-cluster
Status ACTIVE
Task definition first-run-task-definition:1
Service type REPLICA
Launch type FARGATE
Service role AWSServiceRoleForECS
Created By arn:aws:iam::702306393152:role/Admin
Desired count 1
Pending count 0
Running count 1
Details
Tasks
Events
Auto Scaling
Deployments
Metrics
Tags
Logs
Load Balancing
Target Group Name
Container Name
Container Port
EC2Co-Defau-7KT7DL2Z8MOW
sample-app
80
Network Access
Health check grace period 0
Allowed VPC vpc-0ee9af62e5e550290
Allowed subnets subnet-02dbeea54da600e2b,subnet-04d7c111f8a5ac6d0
Security groups* sg-0751d60ab53cbbc18
Auto-assign public IP ENABLED


EC2 > Target groups
Target groups (1) Info
Actions
Create target group
search: arn:aws:elasticloadbalancing:us-east-1:702306393152:targetgroup/EC2Co-Defau-7KT7DL2Z8MOW/fedc2790079d54a7
Clear filters
Name
ARN
Port
Protocol
Target type
Load balancer
VPC ID
EC2Co-Defau-7KT7DL2Z8MOW
80
HTTP
IP
EC2Co-EcsEl-A7GQKEJBDN77
vpc-0ee9af62e5e550290
0 target groups selected
Select a target group above.







Post-Experiments Exercise

Deploy Docker Containers on Amazon ECS (YouTube Video):
https://www.youtube.com/watch?v=9hCZYBs0h9I
Watch video

Extended Theory: (to be written in hand)

1. How is Amazon ECS different from AWS Elastic Beanstalk?

Results/Calculations/Observations:
Fill the following observation tables-(to be written in hand)
Amazon Aurora Vs RDS

Sr. No	**Questions**	**Yes/No**	**Justify**
1.	Does Amazon ECS support any other container types?		
2.	Does Amazon ECS support batch jobs?		
3.	Can user use their own Amazon Machine Image (AMI) on ECS?		

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:
[1] [Online]https://ahireharshal.com/cloud/aws-ecs/
[2] [Online]https://docs.docker.com/cloud/ecs-integration/